

Polarization

(ii) Circularly polarized

$$\vec{E}_1 = E_0 \hat{i} \exp(jky - \omega t)$$

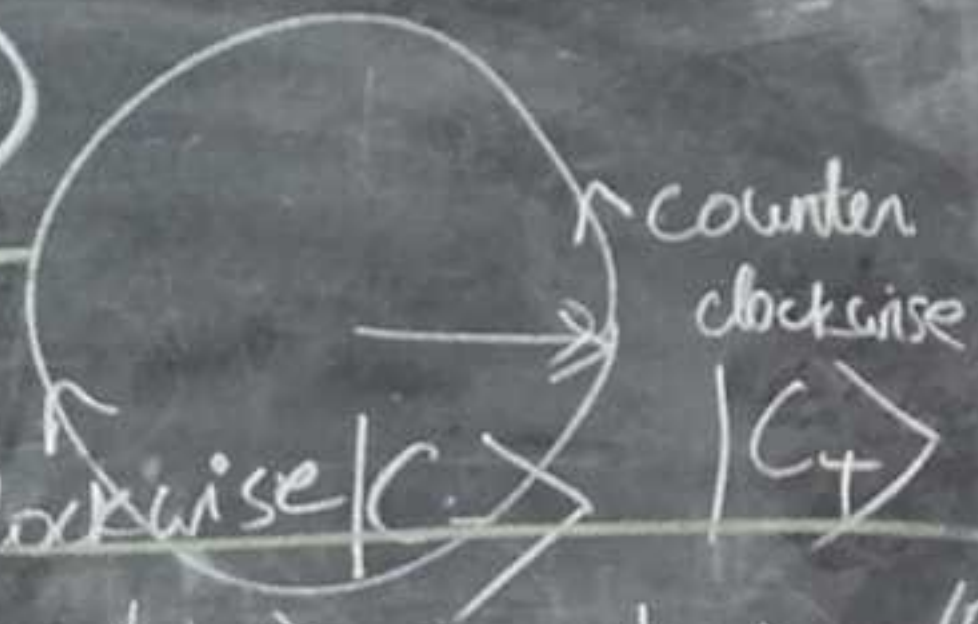
$$\vec{E}_2 = E_0 \hat{j} \exp(jky - \omega t + \frac{\pi}{2})$$

$$|C_{\pm}\rangle = \frac{1}{\sqrt{2}} E_0 (\hat{i} \pm j \hat{j}) \exp(jky - \omega t)$$

$$= \frac{1}{\sqrt{2}} |H\rangle \pm \frac{j}{\sqrt{2}} |V\rangle$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \pm j \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$|C_{\pm}\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm j \end{pmatrix}$$



$$|H\rangle \Rightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |V\rangle \Rightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\vec{E}(y,t) = E_0 \hat{k}$$

$$\vec{E}(y,t) = E_0 \hat{i} \exp(jky - \omega t) \exp(jky - \omega t)$$

where $\alpha, \beta \in \mathbb{C}$

$$|P\rangle = \alpha |H\rangle + \beta |V\rangle$$

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \pm \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm 1 \end{pmatrix}$$

$$= \frac{1}{\sqrt{2}} |H\rangle \pm \frac{1}{\sqrt{2}} |V\rangle$$

$$= \frac{1}{\sqrt{2}} (|H\rangle \pm |V\rangle)$$

$$= \frac{1}{\sqrt{2}} (|H\rangle \pm |V\rangle)$$



$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm 1 \end{pmatrix}$$

Interference

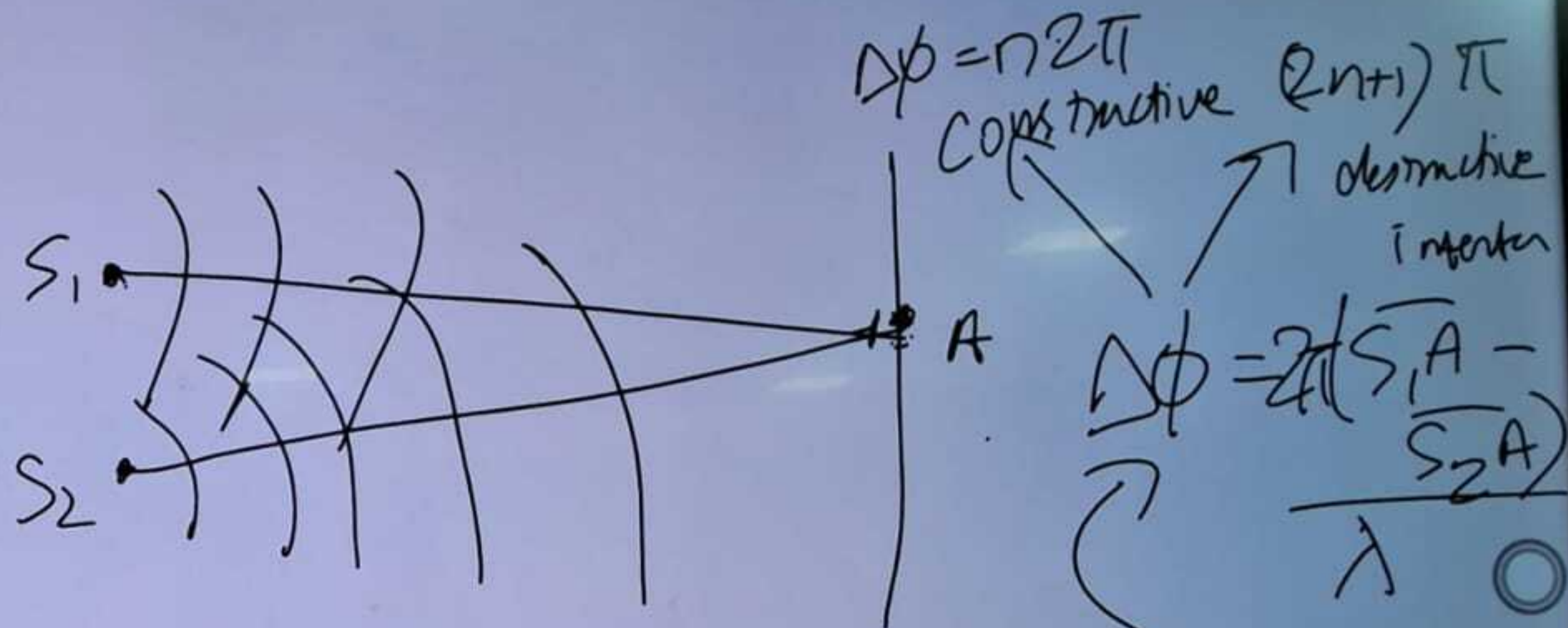
Coherence

Spatially

Temporal



$\frac{x}{t}$



Optical path difference.

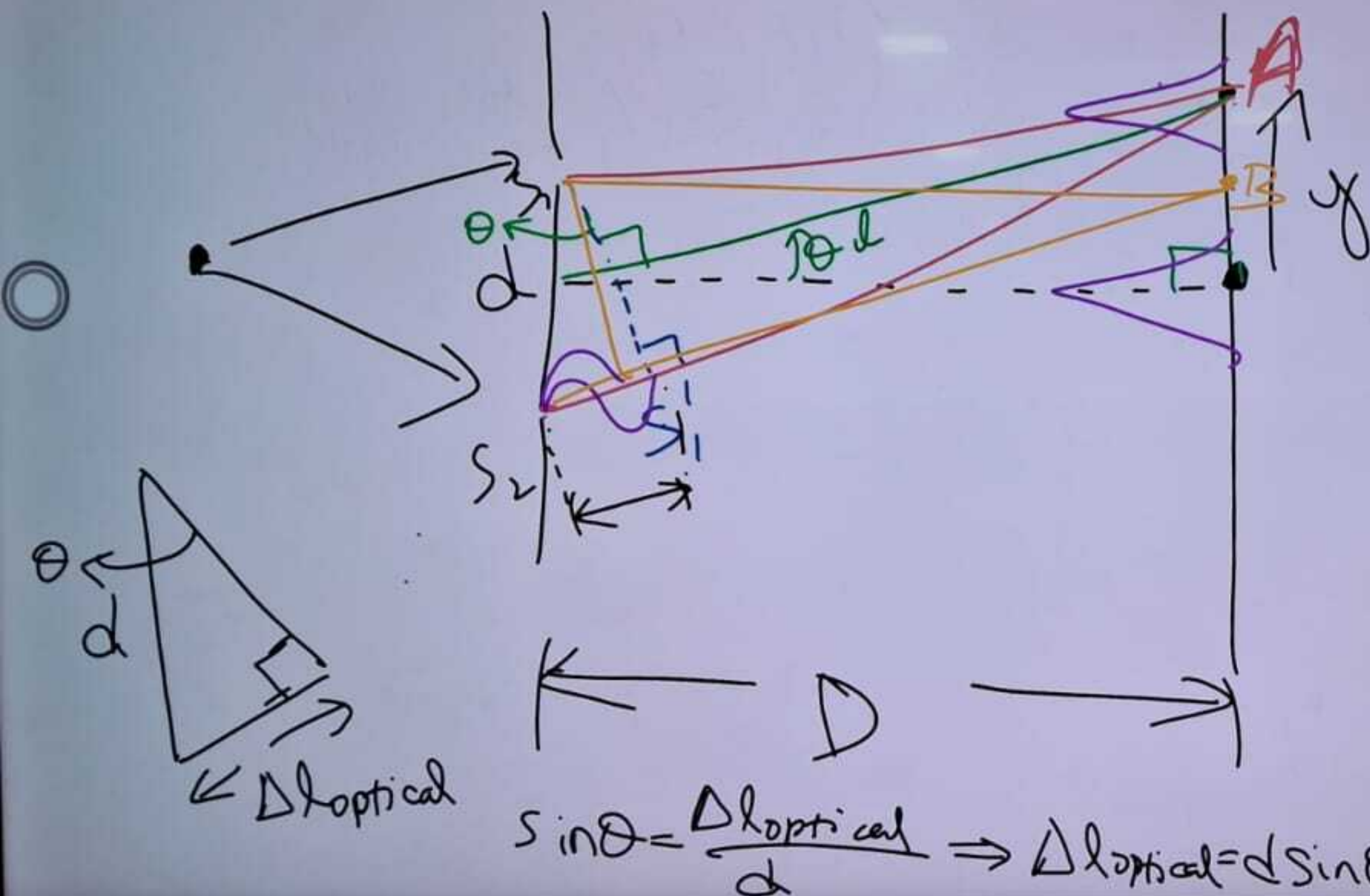
$$\Delta\phi \propto S_1A - S_2A$$

$$\Delta\phi(x)$$

$$\Delta\phi(t)$$

Coherence length
time

Young's double slit



Aim: $I(y)$.

Optical path difference.

$$= |S_1A - S_2A| \quad \text{--- (A)}$$

Approximation: $D \gg d$

$$\Rightarrow S_1A \sim l \quad \text{--- (B)}$$

$$S_2A \sim l$$

$\therefore$ (A) & (B) $\Rightarrow$

$$\Delta l_{\text{optical}} = \overline{S_2S'_1} = d \sin \theta \quad \text{--- (C)}$$

$$(2) \quad I(y) = \langle |\vec{E}|^2 \rangle$$

$$\vec{E}(y) = \vec{E}_1(y) + \vec{E}_2(y)$$

$$I = E_0^2 \left[(1 + \cos \Delta\phi(y))^2 + \sin^2 \Delta\phi(y) \right]$$

From (B), $|\vec{E}_1(y)| \cong |\vec{E}_2(y)| = E_0$

$$\Rightarrow \vec{E}(y) = (E_0 + E_0 \exp(j \Delta\phi(y))) \exp(j \omega t)$$

$$= (E_0(1 + \exp(j \Delta\phi(y)))) \exp(j \omega t)$$

$$= E_0 \left[(1 + \cos \Delta\phi(y)) + j \sin \Delta\phi(y) \right] \exp(j \omega t)$$

$$= E_0^2 (2 + 2 \cos \Delta\phi(y))$$

$$\therefore I(y) = 2 E_0^2 (1 + \cos \Delta\phi(y))$$

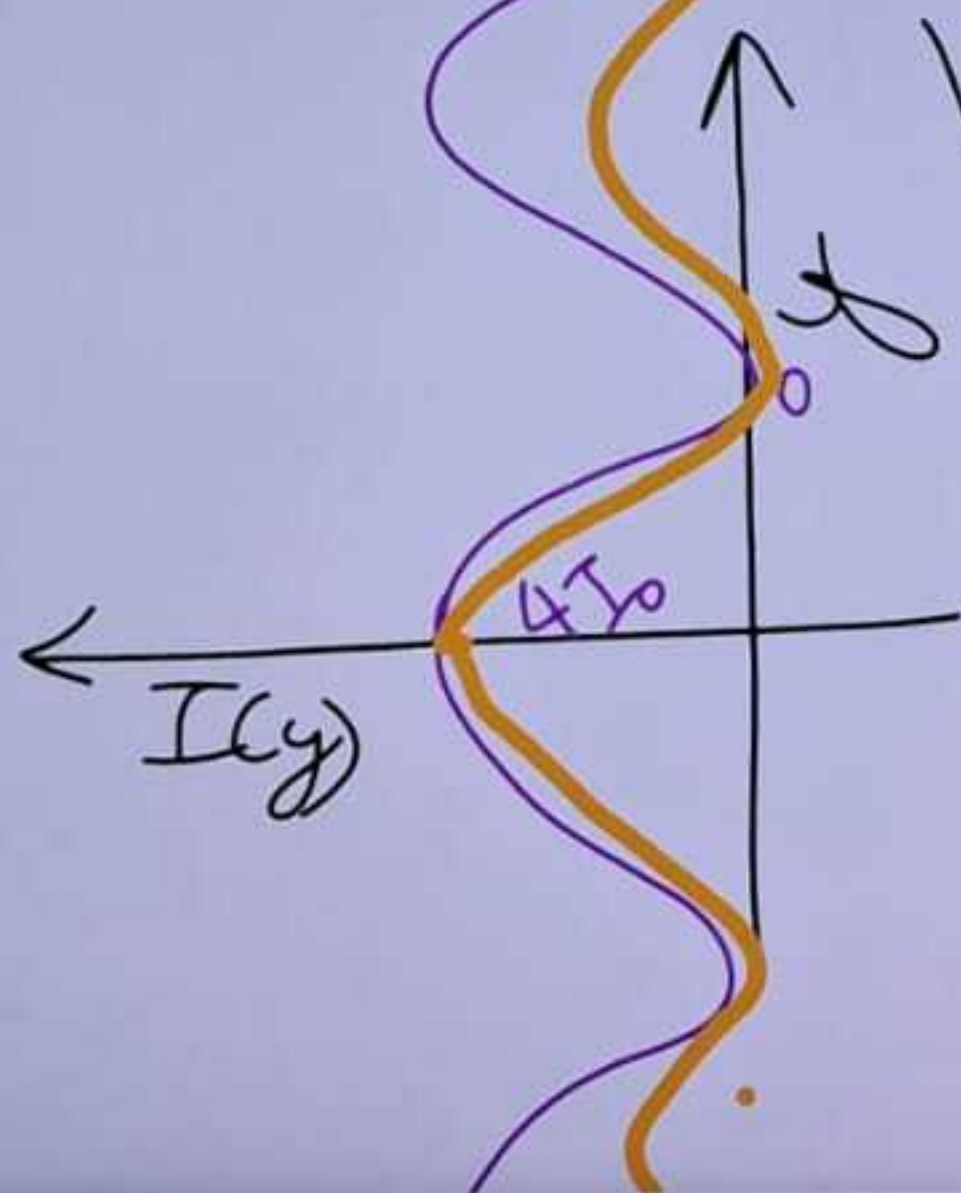
$$I(y) = \begin{cases} 4E_0^2 \\ = 4I_0 \\ 0 \end{cases}$$

$$I(y) = 2I_0 \left(1 + \cos \Delta\phi(y) \right)$$

$$= 2I_0 \left(1 + \cos \frac{2\pi dy}{D\lambda} \right)$$

$$\Delta\phi(y) = \frac{2\pi dy}{D\lambda}$$

$$1 + \cos \theta = 2 \cos^2 \frac{\theta}{2}$$



$$I(y) = 4I_0 \cos^2 \left(\frac{\pi dy}{D\lambda} \right)$$

From (C) & (D) $\Rightarrow \phi = 2\pi$ & $y = \beta$

$$\Delta l_{\text{optical}} = \frac{dy}{D}$$

$$\Delta \phi = \frac{2\pi dy}{D\lambda}$$

$$\Rightarrow 2\pi = \frac{2\pi d\beta}{D\lambda}$$

$$\Rightarrow d = \frac{\lambda D}{\beta}$$

$$\text{But } \tan \theta = \frac{y}{D}$$

Approximation: $y \ll D$

So that $\theta \ll 1$

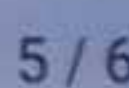
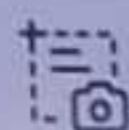
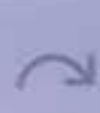
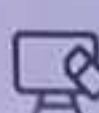
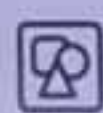
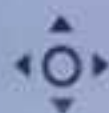
$$\Rightarrow \tan \theta = \theta \Rightarrow \theta = \frac{y}{D}$$

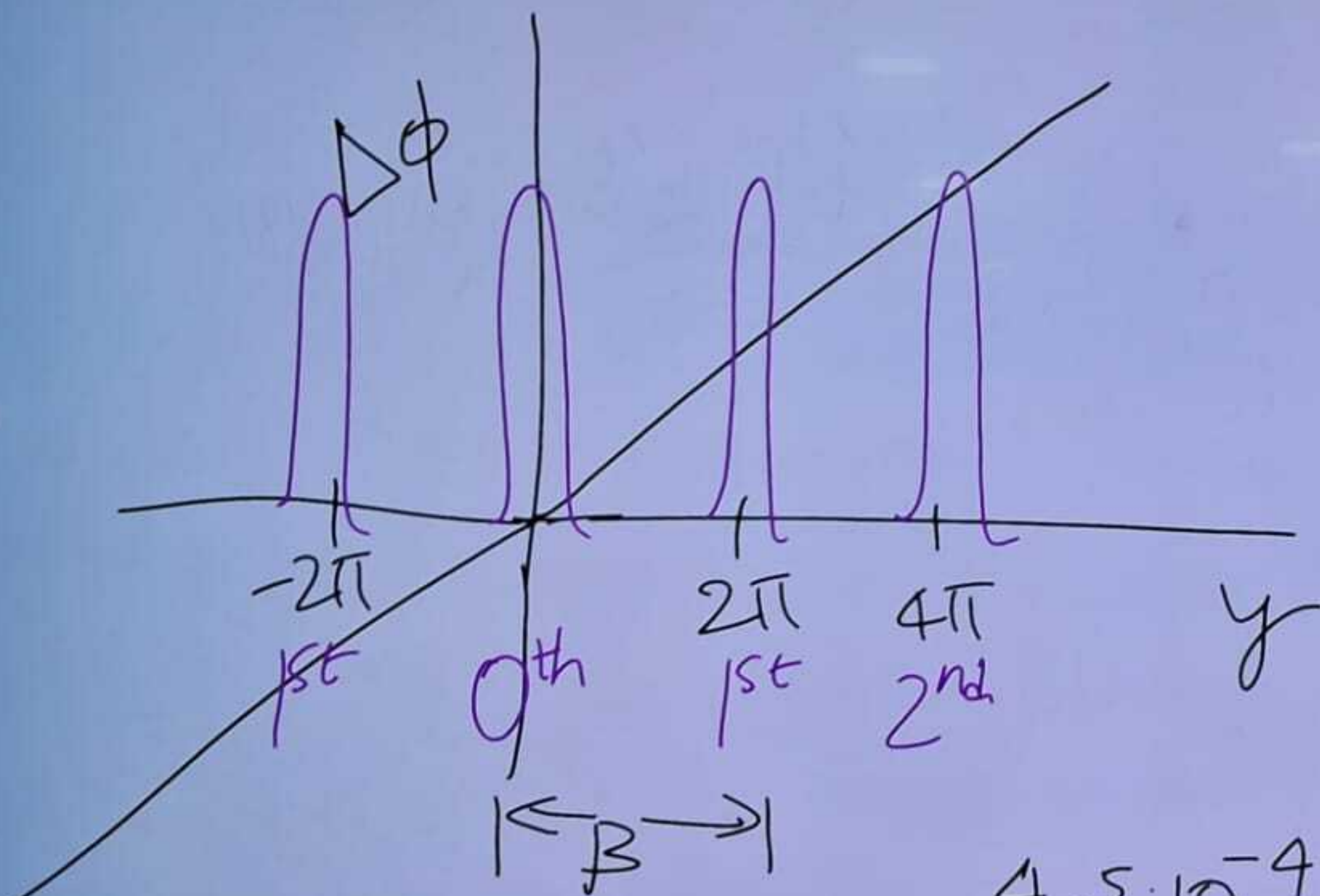
$$\Rightarrow \sin \theta = \theta$$

$\Rightarrow \tan x = x$ when $x \ll 1$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \frac{1}{\cos x} = \lim_{x \rightarrow 0} \frac{\tan x}{x} \quad \&\& \quad \sin x = x \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

when $x \ll 1$





$$\lambda = 5890 \text{ \AA}$$

$$D = 75 \text{ cm}$$

$$\beta = 0.9424 \text{ mm}$$

$$d = ?$$

$$d = \frac{\lambda D}{\beta}$$

$$4.5 \cdot 10^{-4} \text{ m} = \frac{5.89 \cdot 10^{-7} \cdot 7.5 \cdot 10^{-1}}{9.424 \cdot 10^{-4}}$$

$$= 0.045 \text{ cm} \checkmark$$